Fiscal Year:	FY 2019	**Task Last Updated:**	FY 11/03/2018
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John Ph.D. (FM Global)		
Grant/Contract No.:	NNX15AD06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same Principal Investigator (PI) (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station (ISS). The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows: 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	One journal paper was published in 2018 and a second one was accepted. Three conference papers, posters, and abstracts were published and presented. Akshit Markan defended his Ph.D. dissertation and graduated. He has started a position at Whirlpool Inc. Parham Dehghani started his Ph.D. Eric Auth started his M.S. Alex Snegirev and his team continued to perform Fluent simulations of the BRE flames in microgravity. James Quintiere and Parham Dehghani visited NASA Glenn Research Center (GRC) to learn about the flight hardware and operations. Work continued in finalizing the ISS test matrix. We prepared plots of 5 s drop test data. A spreadsheet was created that includes calculations of estimated post-test O2, CO2, and H2O mole fraction and oxidizer bottle pressure. Scrubbing probably will not be needed. A test point parameter spreadsheet was created and modified for simulated tests at NASA GRC. We performed calibrations of heat flux gauges SN175681 and SN175682 using the transfer standard from NIST (National Institute of Standards & Technology). We calibrated the top-plate calorimeter for the 25 mm and 50 mm BRE-ISS duplicate burners. Uniform heat fluxes are applied with step-function changes. The plate temperature was logged. A control-volume analysis allows the rate of temperature change to be converted to the heat flux. Three fitting constants were optimized in Matlab. Various smoothing options were tested. We designed an igniter similar to that on ISS to study ignition conditions at different igniter locations/orientations. We created a Kanthal A1 igniter tip to mimic the one used by NASA and developed a Matlab code to predict igniter temperature with fuel flowing in normal gravity. The optimum radius for the ignitor position was tested. The ISS radiometer diagnostics were reviewed. A similar system was configured for testing at University of Maryland. We stripped the paint and repainted (with Rustoleum paint) several of our burners and heat flux gauges.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Abstracts for Journals and Proceedings	Markan A, Baum HR, Sunderland PB, Quintiere JG, de Ris JL. "Transient Ellipsoidal Combustion Model for the Burning Rate Emulator." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Articles in Peer-reviewed Journals	Markan A, Sunderland PB, Quintiere JG, de Ris JL, Stocker DP, Baum HR. "A Burning Rate Emulator (BRE) for study of condensed fuel burning in microgravity." Combustion and Flame. 2018 Jun;192:272-82. https://doi.org/10.1016/j.combustflame.2018.01.044 , Jun-2018

Papers from Meeting Proceedings	Markan A, Baum HR, Sunderland PB, Quintiere JG, de Ris JL. "Ellipsoidal Combustion Model for a Porous Burner in Microgravity." 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. , Jul-2018
Papers from Meeting Proceedings	Markan A, Baum HR, Sunderland PB, Quintiere JG, de Ris JL. "Transient Ellipsoidal Flames in Microgravity." 2018 Eastern States Section of the Combustion Institute Spring Technical Meeting, State College PA, March 4-7, 2018. 2018 Eastern States Section of the Combustion Institute Spring Technical Meeting, State College PA, March 4-7, 2018. 6 pp. , Mar-2018